The margin–sensitivity coupling: resolved results, the one controlling quantity, and the remaining open path

Working note (theory worked out by hand; agents used only for literature/refutation checks)

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Abstract

This note isolates the "self-limiting coupling" theory behind the paper's margin–sensitivity discussion. Earlier drafts stated four conjectures (A–D); this version records what is now settled. The single controlling quantity is the *spherical log-margin gradient* $\|\nabla_{\mathbb{S}} \log m(u)\|$, equivalently $\tan\angle(\nabla M(x), x)$: the upper envelope on $\|\nabla M\|$ holds iff this is bounded, and the same quantity controls the certificate's looseness. We give the exact identity (§4), a sharp linear case, a conditional bound under global max-margin, an explicit counterexample showing no universal constant exists, the corrected lockstep statement (§5), the corrected local capacity statement (§6), and the bracket-condition-number result (§7). The implicit-bias envelope at the gradient-descent solution is closed (§8) by replacing the global margin with the achieved normalized margin γ_{IB}, which recovers the global statement when the limit is optimal and degrades gracefully to a KKT point otherwise. Everything labelled Proposition/Theorem here is proved.

1 Setup

Let $f : \mathbb{R}^d \to \mathbb{R}^k$ be a classifier with logits $f_1, \dots, f_k$ and, for a labelled point (x, y), signed margin $M(x) = f_y(x) - \max_{j \neq y} f_j(x)$, so $M > 0$ iff f is correct. Write $\eta = \mathbb{E}[M]$ over correct points and $L = \mathbb{E}\|\nabla M\|$ for the mean input-gradient norm (the local sensitivity; norm dual to the threat model). The first-order certificate is $r_2 \geq M(x)/\|\nabla M(x)\|_2$ (Hein–Andriushchenko 2017; Tsuzuku et al. 2018), so η/L is a robustness surrogate. The empirical finding is that η/L *diagnoses* robustness across architectures yet *training* it does not buy robustness; the question is what, precisely, is true behind that.

2 The radial envelope (Euler)

Proposition 1 (Homogeneity lower envelope)**.** *Let f be bias-free with degree-one positively homogeneous activations (ReLU, leaky-ReLU, absolute value, maxout). Then M is degree-one positively homogeneous, $\langle \nabla M(x), x\rangle = M(x)$ (Euler) holds a.e., and hence*

$$\|\nabla M(x)\|_2 \;\geq\; \frac{M(x)}{\|x\|_2}, \qquad \|\nabla M(x)\|_1 \;\geq\; \frac{M(x)}{\|x\|_\infty}, \qquad \eta/L \leq \sup_i \|x_i\|_2.$$

This is exact and standard (Euler's identity plus Cauchy–Schwarz / Hölder). The content is only the reading that the margin gradient has an architecture-imposed *radial* floor. It controls the radial component of the gradient and nothing else, which is exactly the limitation made precise in §4.

3 Why the "impossibility" reading of the coupling is wrong

An earlier draft read Proposition 1 as forbidding the training of η/L. It does not.

(W1) *Wrong direction.* To raise $\eta/L = M/\|\nabla M\|$ one wants $\|\nabla M\|$ small. Proposition 1 is a *lower* bound on $\|\nabla M\|$, i.e. an *upper* cap $\eta/L \leq \|x\|$; an upper cap above the operating value cannot forbid increasing the ratio.

(W2) *Loose at the operating point.* Empirically $\|\nabla M\|_2 \approx 26\, M/\|x\|_2$, an order of magnitude of slack.

(W3) *Hypotheses fail.* Euler needs bias-free, exactly degree-1 homogeneous; the trained models carry biases and batch normalization.

(W4) *Static vs. dynamic.* The empirical obstruction is a property of the training path, not of a static inequality.

What survives unconditionally is the scale degeneracy.

Lemma 1 (Scale degeneracy of the ratio)**.** *$M(x)/\|\nabla M(x)\|$ is invariant under $f \mapsto cf$ $(c > 0)$. Hence "maximize η/L" (equivalently penalize $\|\nabla M\|/M$) is blind to absolute margin and admits the constant classifier $\nabla M \equiv 0$ as a global optimizer; direct ratio maximization is ill-posed. (Tsuzuku et al. 2018, App. G.2.)*

4 The exact spherical envelope (resolves A)

The radial floor controls only the radial gradient. Writing the gradient in polar form isolates the missing piece exactly.

Proposition 2 (Radial–tangential split of the margin gradient)**.** *Let $M : \mathbb{R}^d \setminus \{0\} \to \mathbb{R}$ be C^1 and positively 1-homogeneous. For $x \neq 0$ set $r = \|x\|_2$, $u = x/r$, and $m(u) = M(u)$, so $M(x) = r\, m(u)$. Let $\nabla_{\mathbb{S}} m(u)$ be the spherical gradient (the tangential part). Then*

$$\nabla M(x) = m(u)\, u + \nabla_{\mathbb{S}} m(u), \qquad u \perp \nabla_{\mathbb{S}} m(u),$$

hence

$$\|\nabla M(x)\|_2^2 = \Big(\frac{M(x)}{\|x\|_2}\Big)^2 + \|\nabla_{\mathbb{S}} m(u)\|_2^2,$$

and, when $M(x) > 0$,

$$\boxed{\|\nabla M(x)\|_2 = \frac{M(x)}{\|x\|_2}\sqrt{1 + \|\nabla_{\mathbb{S}} \log m(u)\|_2^2}}\,.$$

Consequently the two-sided envelope $\frac{M(x_i)}{\|x_i\|_2} \leq \|\nabla M(x_i)\|_2 \leq \kappa\, \frac{M(x_i)}{\|x_i\|_2}$ holds at points x_i iff *$\|\nabla_{\mathbb{S}} \log m(u_i)\|_2 \leq \sqrt{\kappa^2 - 1}$, with $u_i = x_i/\|x_i\|_2$. Equivalently $\kappa_i = \sec \angle(\nabla M(x_i), x_i)$.*

Proof. Decompose a perturbation $h = a\, u + z$ with $a = \langle h, u\rangle$, $z = h - au \perp u$. Then $Dr(x)[h] = \langle u, h\rangle = a$ and $Du(x)[h] = (h - au)/r = z/r$ (tangent to $\mathbb{S}$). From $M(x) = r\, m(u)$, $DM(x)[h] = a\, m(u) + r\, Dm(u)[z/r] = a\, m(u) + \langle \nabla_{\mathbb{S}} m(u), z\rangle$. Setting $g = m(u)u + \nabla_{\mathbb{S}} m(u)$ and using $\langle u, z\rangle = 0$, $\langle \nabla_{\mathbb{S}} m(u), u\rangle = 0$ gives $\langle g, h\rangle = DM(x)[h]$ for all h, so $\nabla M(x) = g$. Orthogonality of the two pieces gives the Pythagorean norm identity; $m(u) = M(x)/\|x\|_2$ and $\nabla_{\mathbb{S}} \log m = \nabla_{\mathbb{S}} m/m$ give the boxed form. The envelope equivalence is immediate. □

The controlling quantity. Euler fixes the radial component $m(u) = M/\|x\|$; the entire upper envelope is the *tangential* component $\|\nabla_{\mathbb{S}} \log m(u)\| = \tan\angle(\nabla M, x)$. "Can the coupling be tightened" is exactly "is the spherical log-margin gradient bounded at the support directions." Homogeneity alone gives no such bound (it constrains only the radial part), so A is not provable from homogeneity; it requires an implicit-bias input.

4.1 Sharp linear case

Proposition 3 (Exact for linear max-margin)**.** *For binary labels $y_i \in \{\pm 1\}$ and a linearly separable dataset, let $M_w(x) = y\, w^\top x$ and let $w^\star = \arg\min \|w\|_2$ s.t. $y_i w^\top x_i \geq 1$ be the hard-margin solution, with geometric margin $\gamma = \sup_{\|w\|_2 \leq 1} \min_i y_i w^\top x_i$. Then $\|w^\star\|_2 = 1/\gamma$, and at every support point ($y_i (w^\star)^\top x_i = 1$),*

$$\frac{\|\nabla_x M_{w^\star}(x_i)\|_2}{M_{w^\star}(x_i)/\|x_i\|_2} = \frac{\|x_i\|_2}{\gamma}, \qquad \|\nabla_{\mathbb{S}} \log m(u_i)\|_2 = \sqrt{\left(\tfrac{\|x_i\|_2}{\gamma}\right)^2 - 1}.$$

So $\kappa_i = \|x_i\|_2/\gamma$ exactly: the envelope constant is the data condition number (radius-to-margin), and it is sharp.

Proof. $\nabla_x M_w = y_i w$ so $\|\nabla_x M_w\|_2 = \|w\|_2$. The norm–margin duality $\|w^\star\|_2 = 1/\gamma$ is the standard rescaling argument (normalize any feasible w to unit norm to lower-bound γ; rescale a near-optimal unit v to feasibility to upper-bound). At a support point $M_{w^\star}(x_i) = 1$, so $\|\nabla M\|_2/(M/\|x_i\|_2) = \|w^\star\|_2 \|x_i\|_2 = \|x_i\|_2/\gamma$; substitute into Proposition 2. □

4.2 Conditional bound under global max-margin

Theorem 1 (Complexity-controlled max-margin envelope)**.** *Let the model be positively homogeneous of degree $D > 0$ in its parameters, $M_{a\theta}(x_i) = a^D M_\theta(x_i)$, and let $C(\theta) \geq 0$ be a complexity with the same homogeneity, $C(a\theta) = a^D C(\theta)$, that controls the local input gradient, $\|\nabla_x M_\theta(x_i)\|_2 \leq B_i C(\theta)$ for all i. Define $\gamma_C = \sup_{C(\theta)\leq 1} \min_i M_\theta(x_i) \in (0, \infty)$ and let $\theta^\star$ solve $\min_\theta C(\theta)$ s.t. $M_\theta(x_i) \geq 1$. Then $C(\theta^\star) = 1/\gamma_C$ and, at every support point,*

$$\|\nabla_x M_{\theta^\star}(x_i)\|_2 \leq \frac{B_i}{\gamma_C} = \frac{B_i \|x_i\|_2}{\gamma_C} \cdot \frac{M_{\theta^\star}(x_i)}{\|x_i\|_2}, \qquad \textit{i.e. } \kappa_i \leq \frac{B_i \|x_i\|_2}{\gamma_C}.$$

If moreover $M_{\theta^\star}$ is 1-homogeneous in x, then $\|\nabla_{\mathbb{S}} \log m(u_i)\|_2 \leq \sqrt{(B_i \|x_i\|_2/\gamma_C)^2 - 1}$.

Proof. $C(\theta^\star) = 1/\gamma_C$ is the homogeneous norm–margin duality: any feasible θ rescaled by $C(\theta)^{-1/D}$ has unit complexity and margins $\geq C(\theta)^{-1}$, so $\gamma_C \geq C(\theta)^{-1}$, giving $C(\theta) \geq 1/\gamma_C$; conversely a near-optimal unit-complexity θ_ε rescaled to feasibility has complexity $\leq 1/(\gamma_C - \varepsilon)$. Substitute $C(\theta^\star) = 1/\gamma_C$ into the gradient-control hypothesis and use $M_{\theta^\star}(x_i) = 1$ at support; the spherical form follows from Proposition 2. □

Corollary 1 (Bias-free depth-D ReLU networks)**.** *For a bias-free depth-D network with 1-Lipschitz, 1-homogeneous activations and $C(\theta) = (\|\theta\|_2/\sqrt{D})^D$ (with $\|\theta\|_2^2 = \sum_\ell \|W_\ell\|_F^2$), the input-gradient bound holds with $B_i = \sqrt{2}$: at a differentiability point with a unique rival class $j^\star$, $\nabla_x M = J_f^\top (e_y - e_{j^\star})$, so $\|\nabla_x M\|_2 \leq \sqrt{2}\, \|J_f\|_{2\to 2} \leq \sqrt{2} \prod_\ell \|W_\ell\|_2 \leq \sqrt{2} \prod_\ell \|W_\ell\|_F \leq \sqrt{2}\, (\|\theta\|_2^2/D)^{D/2} = \sqrt{2}\, C(\theta)$, the last step by AM–GM. Hence, at the min-norm (global max-margin) interpolant, $\kappa_i \leq \sqrt{2}\, \|x_i\|_2/\gamma_{\mathrm{net}}$.*

4.3 No universal constant (the counterexample)

Proposition 4 (The envelope constant must depend on data geometry)**.** *There is no universal C with $\|\nabla_{\mathbb{S}} \log m(u_i)\|_2 \le C$ at the max-margin solution, even for linear classifiers. Fix $A \ge 1$ in $\mathbb{R}^2$ and take $(e_1, +1), (-e_1, -1), (e_2/A, +1), (-e_2/A, -1)$. The hard-margin solution is $w^\star = (1, A)$ (constraints force $w_1 \ge 1$, $w_2 \ge A$, minimized at equality), so at $x_1 = e_1$: $M = 1$, $\nabla_x M = (1, A)$, radial part $(1, 0)$, tangential part $(0, A)$, hence $\|\nabla_{\mathbb{S}} \log m(e_1)\|_2 = A$ and $\kappa = \sqrt{1 + A^2} = \|x_1\|_2/\gamma \to \infty$.*

This realizes the spherical blow-up at a genuine max-margin solution; agent D (§7) exhibits the same blow-up for a bias-free degree-1 *ReLU* net ($\kappa \approx 21$ at maximal geometric margin), so the dependence on the data condition number is real for the architecture, not an artifact of linearity.

5 The lockstep, corrected (resolves B)

The conjectured inequality $\frac{d\eta}{d(\log L)} \ge c\eta$ is mis-signed for a no-go: with $d \log L/d\lambda < 0$ it only lower-bounds margin decay and permits $d \log \eta/d\lambda = 0$ (sensitivity drops at fixed margin), the opposite of an obstruction. The correct statements:

Proposition 5 (Exact lockstep on a scale orbit)**.** *Fix a margin function $\bar{M}$ on $x_1, \ldots, x_n$ and let $M_\lambda = a(\lambda)\bar{M}$ with $a(\lambda) > 0$. Then $\eta(\lambda) = a(\lambda)\bar{\eta}$, $L(\lambda) = a(\lambda)\bar{L}$, so $\eta/L \equiv \bar{\eta}/\bar{L}$ is constant and, if $a' \ne 0$, $d \log \eta/d \log L = 1$. A first-order penalty cannot change η/L by mere rescaling; it can only change it by changing the classifier's* shape*.*

Proof. $M_\lambda(x_i) = a(\lambda)\bar{M}_i$ and $\nabla_x M_\lambda = a(\lambda)\nabla_x \bar{M}$ give $\eta(\lambda) = a\bar{\eta}$, $L(\lambda) = a\bar{L}$; the ratio cancels a and the log-derivatives coincide. □

Proposition 6 (Implicit-function characterization; no lockstep in general)**.** *Let θ_λ minimize $\mathcal{L}(\theta) + \lambda R(\theta)$ with $R = \mathbb{E}\|\nabla_x M\|_q$ and $H_\lambda = \nabla^2 \mathcal{L} + \lambda \nabla^2 R$. Then $\dot{\theta}_\lambda = -H_\lambda^{-1} \nabla R$, $dL/d\lambda = -\langle \nabla R, H_\lambda^{-1} \nabla R\rangle \le 0$ (the penalty reduces sensitivity for free), and $d(\eta/L)/d\lambda = -\langle \nabla(\eta/L), H_\lambda^{-1} \nabla R\rangle$ is an alignment/operating-point quantity, not a fixed sign. In the linear case along the ridge path $w(t) = (\Sigma + tI)^{-1} b$, $d \log \eta/d \log L = m_2^2/(m_1 m_3) \le 1$ ($m_k = b^\top A^{-k} b$, $A = \Sigma + tI$; Cauchy–Schwarz), so η/L is* non-decreasing*: the penalty raises the surrogate. At a ratio-critical (robust-optimal) operating point $d(\eta/L)/d\lambda = 0$ to first order, which is where an adversarially-trained network sits; this is the honest explanation of the four-intervention experiment. The reason raising the surrogate does not raise robustness is that $\eta = \mathbb{E}[M]$ is a weak robustness surrogate (raising it can rotate the classifier away from the robust-optimal direction).*

6 The local capacity ceiling, corrected (resolves C)

As stated C is false without noise: for noiseless $x_i \sim \mathcal{N}(0, I_d)$ with $y_i = \mathrm{sign}(u^\top x_i)$, the interpolant $g(x) = u^\top x$ has $L = 1$, $\eta = \sqrt{2/\pi}$, so $\eta/L = \Theta(1) \gg \tilde{O}(\sqrt{p/(nd)})$; a constant classifier gives $L = 0$, $\eta/L = \infty$. (This matches Bubeck–Sellke 2021, who state both that noise is necessary and that the bound needs the *supremum*, not the average, of $\|\nabla f\|$.) The salvage is a local, noisy-pair statement.

Proposition 7 (Local ceiling from close opposite-label pairs)**.** *Let g be C^1 on a convex set, with signed margins $M_i = y_i g(x_i) > 0$ and $G_i = \|\nabla g(x_i)\|_2$. Suppose the data form m disjoint opposite-label pairs (a_s, b_s) at distances $\rho_s \le \rho_{\max}$, each segment in the domain, with the endpoint-control* *condition* $\int_0^1 \|\nabla g(\gamma_s(t))\|_2\, dt \le \frac{C}{2}(G_{a_s} + G_{b_s})$ *for some* $C \ge 1$. *With* $\eta = \frac{1}{2m}\sum_s(M_{a_s} + M_{b_s})$, $L = \frac{1}{2m}\sum_s(G_{a_s} + G_{b_s})$, *then* $\eta \le \frac{C\rho_{\max}}{2}L$, *i.e.* $\eta/L \le C\rho_{\max}/2$.

Proof. For a pair, $M_a + M_b = g(x_a) - g(x_b) = \int_0^1 \langle \nabla g(\gamma(t)), x_a - x_b\rangle dt \le \rho \int_0^1 \|\nabla g(\gamma(t))\|_2 dt \le \frac{C\rho}{2}(G_a + G_b)$ by FTC, Cauchy–Schwarz, and endpoint control; sum over pairs and divide by $2m$. □

This is the honest local analogue of the law of robustness: a ceiling needs close opposite-label forcing plus no gradient spikes on the connecting segment.

7 The bracket condition number (resolves D)

The certificate looseness is $\kappa_{\text{br}} = L\, r_2/\eta \ge 1$, with $\kappa_{\text{br}} = 1$ iff M is affine on the segment from x to the boundary, so κ_{br} measures failure of local linearity, a curvature quantity, not a margin quantity. D as stated (a bound $g(\mu)$ in the separation margin alone) is false: a bias-free degree-1 ReLU net can have $\kappa_{\text{br}} \approx 21$ at *maximal* geometric margin. Two true statements:

Proposition 8 (Architectural bound; benign curvature)**.** (P1) *For any bias-free degree-*1 *homogeneous net,* $\kappa_{\text{br}} \le \sec\angle(\nabla_x M(x), x)$, *the same envelope constant as Proposition 2; in particular the polynomial* $t \mapsto t^{2n+1}$ *(degree* > 1*) blow-up is excluded (*$\kappa = 2n+1$ *there comes from non-unit homogeneity).* (P2) *If* M *has curvature* β *along the path with* $2\beta\eta \le L^2$ *then* $\kappa_{\text{br}} \le 2/(1 + \sqrt{1 - 2\beta\eta/L^2}) \le 2$ *(a CURE / local-linearity-regularization style bound).*

P1 ties the bracket looseness to the *same* spherical/angular quantity as the envelope, so η/L is loose exactly when the input gradient is tangentially misaligned with the datum; the implicit bias does not certify the benign regime (min-norm interpolants permit localized curvature spikes; this is why curvature/local-linearity regularizers exist).

8 The one controlling quantity, and the resolved implicit-bias gap

Synthesis. A single quantity governs both the envelope and the certificate: the spherical log-margin gradient $\|\nabla_{\mathbb{S}} \log m(u_i)\| = \tan\angle(\nabla M, x_i)$ at support directions, with $\sec\angle = \sqrt{1 + \tan^2}$ equal to the envelope κ (Prop. 2) and an upper bound on the bracket κ_{br} (Prop. 8). It is exactly the data condition number $\|x\|/\gamma$ for linear max-margin (Prop. 3), bounded by $B_i\|x\|/\gamma_C$ under global max-margin with gradient-controlling complexity (Thm. 1, Cor. 1), and unbounded across datasets (Prop. 4). The margin alone never bounds it.

The gap, closed by the achieved margin. Theorem 1 and Corollary 1 were stated for the *global* max-margin interpolant. Gradient descent on homogeneous networks converges only to a *KKT point* of the max-margin program (Lyu–Li 2020, arXiv:1906.05890; Ji–Telgarsky 2020, arXiv:2006.06657), which for ReLU networks need not be a local or global max-margin optimum (Vardi–Shamir–Srebro 2022, arXiv:2110.02732); the linear case is the clean benchmark, where GD converges to the hard-margin SVM direction (Soudry et al. 2018, JMLR). A proof that depends only on "GD reaches a KKT point" therefore cannot deliver the *global*-margin bound. The fix is to replace the global margin γ_C by the normalized margin *actually achieved* by the limiting direction, γ_{IB}; the envelope then holds for the implicit-bias output itself and recovers the global statement when that output is globally optimal.

Theorem 2 (Implicit-bias spherical envelope via the achieved margin)**.** *Let $x_1, \dots, x_n \in \mathbb{R}^d \setminus \{0\}$ and $M_\theta(x_i)$ be the signed margin. Assume* (i) *parameter homogeneity $M_{a\theta}(x_i) = a^p M_\theta(x_i)$ for some $p > 0$;* (ii) *a matching homogeneous complexity $C(a\theta) = a^p C(\theta)$ with $\|\nabla_x M_\theta(x_i)\|_2 \le B_i C(\theta)$ at differentiability points;* (iii) *the limiting gradient-descent direction $\bar\theta$ makes $x \mapsto M_{\bar\theta}(x)$ positively 1-homogeneous; and* (iv) $\min_i M_{\bar\theta}(x_i) > 0$. *Set the achieved normalized margin $\gamma_{\mathrm{IB}} := \min_i M_{\bar\theta}(x_i)/C(\bar\theta)$ (scale-invariant) and normalize $\widehat\theta := (\min_i M_{\bar\theta}(x_i))^{-1/p}\bar\theta$, so $\min_i M_{\widehat\theta}(x_i) = 1$. Then at every support point ($M_{\widehat\theta}(x_i) = 1$),*

$$\|\nabla_x M_{\widehat\theta}(x_i)\|_2 \le \frac{B_i}{\gamma_{\mathrm{IB}}} = \frac{B_i\|x_i\|_2}{\gamma_{\mathrm{IB}}} \cdot \frac{M_{\widehat\theta}(x_i)}{\|x_i\|_2}, \qquad \|\nabla_{\mathbb{S}} \log m(u_i)\|_2 \le \sqrt{\left(\frac{B_i\|x_i\|_2}{\gamma_{\mathrm{IB}}}\right)^2 - 1}.$$

If $B_i \le B$ and $\|x_i\|_2 \le R$ on the support set, $\kappa_i \le BR/\gamma_{\mathrm{IB}}$.

Proof. γ_{IB} is scale-invariant: rescaling $\bar\theta \mapsto a\bar\theta$ multiplies both $\min_i M_{\bar\theta}(x_i)$ and $C(\bar\theta)$ by a^p. With $\alpha = (\min_i M_{\bar\theta}(x_i))^{-1/p}$, parameter homogeneity gives $M_{\widehat\theta}(x_i) = \alpha^p M_{\bar\theta}(x_i) = M_{\bar\theta}(x_i)/\min_j M_{\bar\theta}(x_j)$, so $\min_i M_{\widehat\theta}(x_i) = 1$, and $C(\widehat\theta) = \alpha^p C(\bar\theta) = C(\bar\theta)/\min_i M_{\bar\theta}(x_i) = 1/\gamma_{\mathrm{IB}}$. The gradient-control hypothesis then gives $\|\nabla_x M_{\widehat\theta}(x_i)\|_2 \le B_i C(\widehat\theta) = B_i/\gamma_{\mathrm{IB}}$; at a support point $M_{\widehat\theta}(x_i) = 1$ rewrites this in the displayed form. The spherical version follows by dividing by $m(u_i) = M_{\widehat\theta}(x_i)/\|x_i\|_2$ in Proposition 2. ☐

Corollary 2 (Bias-free ReLU networks)**.** *For a depth-D bias-free network with 1-Lipschitz 1-homogeneous activations and $C(\theta) = (\|\theta\|_2/\sqrt{D})^D$, the gradient-control hypothesis holds with $B_i = \sqrt{2}$ (Corollary 1), so the implicit-bias direction satisfies $\|\nabla_x M_{\widehat\theta}(x_i)\|_2 \le \sqrt{2}/\gamma_{\mathrm{IB}}$ and $\|\nabla_{\mathbb{S}} \log m(u_i)\|_2 \le \sqrt{2\|x_i\|_2^2/\gamma_{\mathrm{IB}}^2 - 1}$ at support points.*

Remark 1 (Why this is the right denominator)**.** When the gradient-descent limit is globally optimal, $\gamma_{\mathrm{IB}} = \gamma_C$ and Theorem 2 reduces to Theorem 1. When it is only a KKT point, the achieved margin $\gamma_{\mathrm{IB}} \le \gamma_C$ is smaller and the bound weakens accordingly, which is exactly the behaviour forced by the Vardi–Shamir–Srebro counterexamples: KKT convergence does not license replacing γ_{IB} by γ_C. The linear case is exact ($\|\nabla_x M_w\|_2 = \|w\|_2$, $C = \|w\|_2$, equality), and Proposition 4 shows no denominator weaker than the solution's data condition number can work. Open residuals: comparing γ_{IB} to γ_C (when GD is globally optimal) is the standard hard implicit-bias question; B in the genuine non-scale nonlinear regime; C in the noisy $p \ll nd$ training-point-average niche; D's explicit GD dataset realizing large κ_{br} at large margin.

Statement	Status	Where
Radial (Euler) lower envelope	proved (standard)	Prop. 1
Ratio-maximization degeneracy	proved (known)	Lem. 1
Exact radial–tangential split	proved (standard polar)	Prop. 2
Linear max-margin: $\kappa = \|x\|/\gamma$	proved, sharp	Prop. 3
Conditional envelope (global max-margin)	proved (adapted duality)	Thm. 1, Cor. 1
No universal constant	proved (counterexample)	Prop. 4
Scale-orbit lockstep / IFT characterization	proved	Prop. 5, 6
Local noisy-pair ceiling	proved	Prop. 7
Bracket $\kappa_{\mathrm{br}} \le \sec\angle$; benign ≤ 2	proved	Prop. 8
Implicit-bias envelope via achieved margin	proved	Thm. 2

Provenance (for honest naming later). Prop. 1, Lem. 1, Prop. 2 are standard/known (Euler identity; Tsuzuku; polar gradient decomposition). Prop. 3 and Thm. 1 are adaptations of the norm–margin duality. Cor. 1 is a direct application of the product-of-spectral-norms Lipschitz bound. Prop. 4 (no universal constant), the identification of the spherical log-gradient as the controlling quantity, and the achieved-margin envelope Thm. 2 (which closes the implicit-bias gap without the unprovable global-optimality assumption) are the genuinely informative contributions.

9 Orbit-complete interpolation has no invariant norm gap (corrects E1 / the lazy-cluster theorem)

The lazy-regime projection argument and its rich-regime variation-norm analogue must be stated with care. If the training set is closed under the group action and labels are constant on orbits, enforcing invariance *cannot* reduce the minimum interpolation norm: the invariant class is a subset of the unconstrained class, while Reynolds averaging maps every feasible unconstrained interpolant to an invariant feasible interpolant of no larger norm, so the two optimal values coincide. Consequently a strict certified-Lipschitz gap cannot come from the same norm certificate on orbit-complete data. **This corrects the draft's lazy-cluster theorem**, which claims a strict gap equal to the orbit-variance $\|(I - \Pi_G) f_{\text{unc}}\|^2$: for a strictly convex norm (the RKHS case) the unconstrained minimum-norm interpolant on orbit-complete data is *already invariant*, so that orbit-variance is exactly zero.

Theorem 3 (No invariant norm gap on orbit-complete data)**.** *Let a compact group G act on the inputs, with induced action $(T_g f)(x) = f(g^{-1}x)$. Let $(\mathcal{F}, \|\cdot\|_{\mathcal{F}})$ be a convex class with a G-invariant convex norm ($\|T_g f\|_{\mathcal{F}} = \|f\|_{\mathcal{F}}$), let μ be normalized Haar measure, and let the Reynolds operator $\Pi_G f := \int_G T_g f \, d\mu(g)$ satisfy $\Pi_G f \in \mathcal{F}$ for $f \in \mathcal{F}$. Let the data $\{(x_i, y_i)\}_{i=1}^n$, $y_i \in \{\pm 1\}$, be* orbit-complete *and* orbit-constant*: for every i and $g \in G$, gx_i is a training point with label y_i. With $\mathcal{F}^G = \{f \in \mathcal{F} : T_g f = f \ \forall g\}$, define the unit-margin minimum-norm values*

$$P_{\text{unc}} = \inf_{f \in \mathcal{F}} \{\|f\|_{\mathcal{F}} : y_i f(x_i) \geq 1 \ \forall i\}, \qquad P_{\text{inv}} = \inf_{f \in \mathcal{F}^G} \{\|f\|_{\mathcal{F}} : y_i f(x_i) \geq 1 \ \forall i\}.$$

Then $P_{\text{unc}} = P_{\text{inv}}$. If an unconstrained minimizer $f_\star$ exists, $\Pi_G f_\star$ is an invariant minimizer; if $\|\cdot\|_{\mathcal{F}}$ is strictly convex, every unconstrained minimizer is already invariant.

Proof. *(1) Π_G is non-expansive.* By Jensen and norm-invariance, $\|\Pi_G f\| \leq \int_G \|T_g f\| \, d\mu = \|f\| \int_G d\mu = \|f\|$ (Haar normalized). *(2) Π_G preserves feasibility and lands in $\mathcal{F}^G$.* If $y_i f(x_i) \geq 1$ for all i, then for each g, $g^{-1}x_i$ is a training point with label y_i, so $y_i (T_g f)(x_i) = y_i f(g^{-1}x_i) \geq 1$; averaging over g gives $y_i (\Pi_G f)(x_i) = \int_G y_i (T_g f)(x_i) \, d\mu \geq 1$. Invariance: $T_h \Pi_G f = \int_G T_{hg} f \, d\mu(g) = \int_G T_k f \, d\mu(k) = \Pi_G f$ by left-invariance of Haar. *(3) Equal values.* $\mathcal{F}^G \subseteq \mathcal{F}$ gives $P_{\text{unc}} \leq P_{\text{inv}}$. For the reverse, take f_ε feasible with $\|f_\varepsilon\| \leq P_{\text{unc}} + \varepsilon$; by (2) $\Pi_G f_\varepsilon$ is invariant feasible and by (1) $\|\Pi_G f_\varepsilon\| \leq P_{\text{unc}} + \varepsilon$, so $P_{\text{inv}} \leq P_{\text{unc}} + \varepsilon$; let $\varepsilon \downarrow 0$. *(4) Self-symmetrization.* If $f_\star$ minimizes, $\Pi_G f_\star$ is invariant feasible with $\|\Pi_G f_\star\| \leq \|f_\star\| = P_{\text{unc}} = P_{\text{inv}}$, hence an invariant minimizer. If the norm is strictly convex and some $f_\star$ had $T_h f_\star \neq f_\star$: both are feasible minimizers of equal norm (each $T_g f_\star$ is feasible since $g^{-1}x_i$ is a same-label training point), so their average is feasible with strictly smaller norm, contradicting optimality; hence $f_\star \in \mathcal{F}^G$. □

Corollary 3 (Variation-norm / rich two-layer)**.** *For an infinite-width two-layer network $f(x) = \int a \, \sigma(w^\top x + b) \, d\rho$ with variation (atomic) norm $\|f\|_{\mathcal{V}} = \inf_\rho \int |a| \, d|\rho|$, if the activation dictionary is closed under the (orthogonal) G-action—for each atom ϕ and g, $T_g \phi$ is again an atom—then*

$\|\cdot\|_{\mathcal{V}}$ *is* G*-invariant and Theorem 3 applies: on orbit-complete data the unconstrained and invariant minimum-variation values coincide. (The norm is not strictly convex, so an invariant optimum* exists *but need not be unique.)*

Corollary 4 (RKHS: orbit-complete data force invariant interpolation)**.** *For an RKHS* $\mathcal{H}_K$ *with a* G*-invariant kernel, the action is an isometry and the norm is strictly convex, so on orbit-complete data the unit-margin minimum-norm interpolant is unique and* G*-invariant. With a certificate* $L_{\mathrm{cert}}(f) = L_\Phi \|f\|_K$*, equal minimizing norms give equal best certified constants: there is* no *strict RKHS-norm or certified-Lipschitz gap from the projection argument on orbit-complete data.*

Remark 2 (What survives, and the correct open problem)**.** The theorem does not say invariance cannot help robustness; only that on orbit-complete data the help cannot be a smaller value of the *same* minimum norm. Three routes remain. (i) On *non*-orbit-complete data, Π_G need not preserve the margin constraints, so the invariant and unconstrained problems genuinely differ. (ii) Even at equal minimum norm, the *realized* local gradient $\mathbb{E}_i\|\nabla_x f(x_i)\|$ is not a norm value and may differ among equal-norm interpolants— this is where a realized η/L advantage lives. (iii) For the non-strictly-convex variation norm the optimum is non-unique, so the rich-regime question is correctly localized: not a smaller variation-norm optimum, but whether the *dynamics select* an invariant / lower-local-gradient optimum among equal-norm interpolants. Two consequences for the paper. First, the empirical lazy-regime cluster advantage (and the synthetic-data smoothness effect) is a *realized*-local-Lipschitz / finite-width trajectory phenomenon, not an infinite-width min-norm gap. Second, on orbit-complete data with a strictly convex (RKHS) norm there is no advantage at all in the exact limit, so the lazy theorem must be restated as no-gap plus realized-Lipschitz localization (this is the corrected E1).